4 **(a)** The gravitational force due to a mass is always attractive. Hence, the external force required to bring a small test mass from infinity to a point in the gravitational field of the mass always acts in the opposite direction to the displacement of the small test mass. The work done per unit mass by the external force is thus negative and gravitational potential is always negative.

The electric force due to a negative or positive source charge on a small positive test charge is attractive or repulsive respectively.

For a negative source charge, the external force required to bring the small (positive) test charge from infinity to a point in the electric field acts in the opposite direction to the displacement of the small test charge. The work done per unit positive charge by the external force is thus negative and electric potential is negative.

For a positive source charge, the external force required to bring the small (positive) test charge from infinity to a point in the electric field acts in the same direction as the displacement of the small test charge. The work done per unit positive charge by the external force is thus positive and electric potential is positive.

(b) **(i)** **1.** The gravitational force on the moon by the planet provides the centripetal force.

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$v = \sqrt{\frac{GM}{r}} = \sqrt{-\phi} \qquad \text{since } \phi = -\frac{GM}{r}$$

when $r = 1.07 \times 10^6 \text{ km} = 10.7 \times 10^8 \text{ m}$, $\phi = -1.2 \times 10^8 \text{ J kg}^{-1}$

$$\begin{aligned} v &= \sqrt{-\phi} \\ &= \sqrt{-(-1.2 \times 10^8)} \\ &= 10954 = 1.10 \times 10^4 \text{ ms}^{-1} \end{aligned}$$

2.

$$\begin{aligned} \text{total energy} &= E_p + E_k \\ &= m\phi + \frac{1}{2}mv^2 \\ &= m\phi - \frac{1}{2}m\phi \\ &= \frac{1}{2}m\phi \\ &= \frac{1}{2}(1.48 \times 10^{23})(-1.2 \times 10^8) \\ &= -8.88 \times 10^{30} \text{ J} \end{aligned}$$

(ii) **1.** at the surface of the planet: $r = 0.7 \times 10^{8}$ m, $\phi_i = -18.1 \times 10^{8}$ J kg^{-1}

when the K.E. of the rock is zero, by conservation of energy,

$$E_i = E_f$$

$$m\phi_i + \frac{1}{2}mv^2 = 0 + m\phi_f$$

$$\phi_f = \phi_i + \frac{1}{2}v^2$$

$$= (-18.1 \times 10^{8}) + \frac{1}{2}(45 \times 10^{3})^2$$

$$= -7.975 \times 10^{8} = -7.98 \times 10^{8} \text{ J kg}^{-1}$$

Since the <u>gravitational potential</u> of the rock is <u>negative when its kinetic energy is zero</u>, the rock cannot reach infinity to escape.

2. From Fig 4.1, when $\phi_f = -7.98 \times 10^{8}$ J kg^{-1}, $r = 1.6 \times 10^{8}$ m

max. distance from surface $= (1.6 - 0.7) \times 10^{8} = 9.0 \times 10^{7}$ m

5 **(a)**